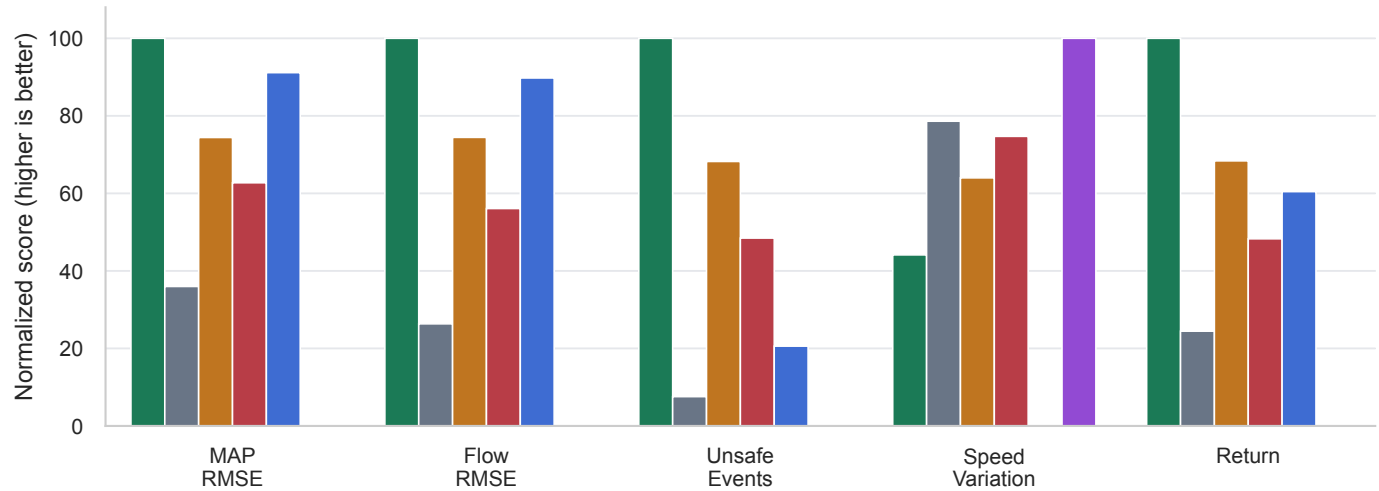


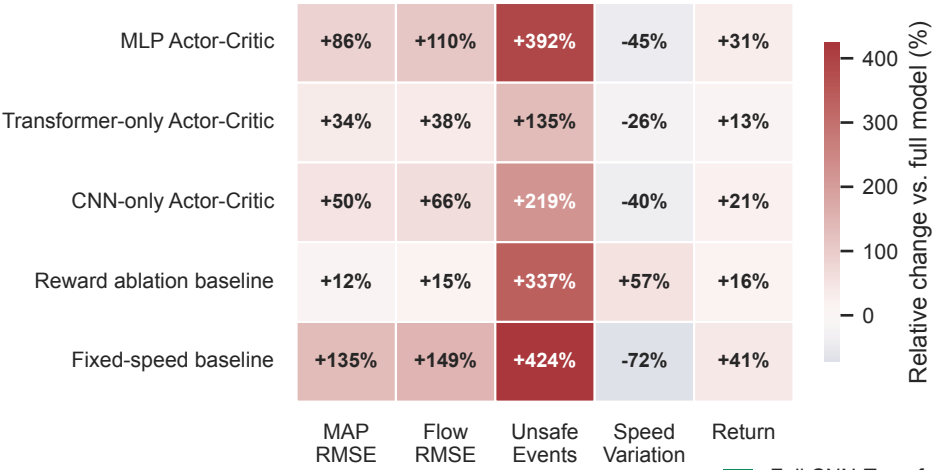
Ablation study of representation modules and safety reward terms

Synthetic pilot data generated from repeated virtual runs; replace the CSV with measured results for final reporting.

A Normalized control quality across ablations



B Performance trade-off relative to full model



Full CNN-Transformer

MLP Actor-Critic

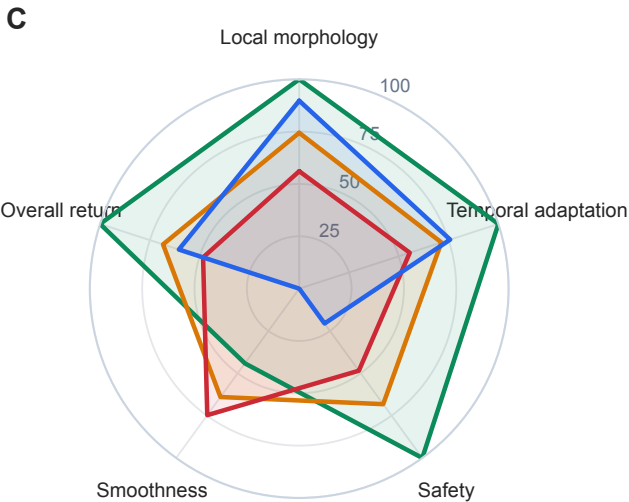
Transformer-only Actor-Critic

CNN-only Actor-Critic

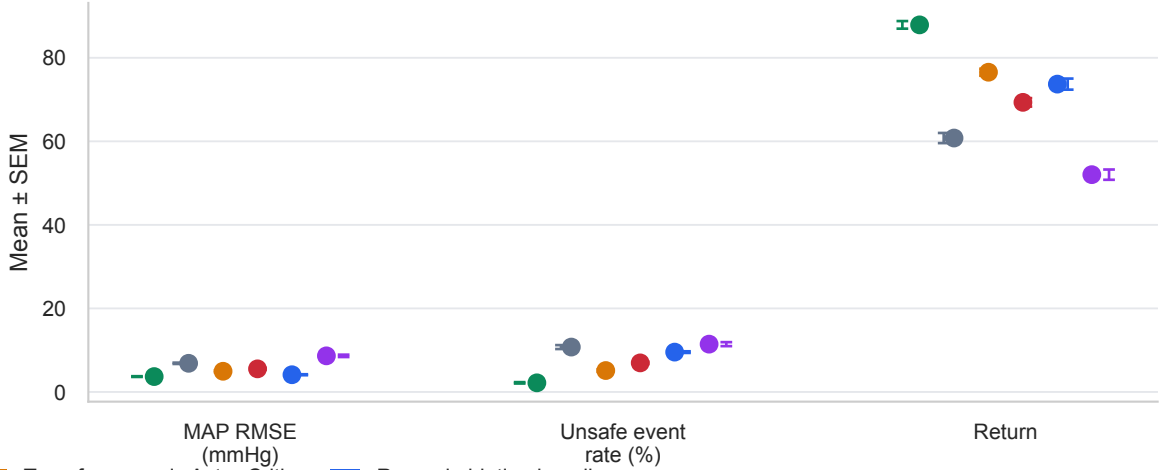
Reward ablation baseline

Fixed-speed baseline

C Module-level contribution profile



D Representative raw metrics



Error bars indicate SEM across virtual runs. Error-type metrics are inverted before normalization in panel A.